

FACTORGSE: SELECTIVE AUTOREGRESSION FOR LOW-LATENCY STREAMING GENERATIVE SPEECH ENHANCEMENT

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ABSTRACT

Autoregression need not be an all-or-nothing choice for codec-based streaming speech enhancement. Temporal feedback can propagate semantic errors, especially with limited look-ahead, whereas fully non-autoregressive (NAR) decoding omits explicit coarse-to-fine dependencies among residual vector quantization codebooks. We introduce FactorGSE, a selective factorization that uses time-non-autoregressive (time-NAR) prediction for a semantically distilled first codebook from bounded degraded speech without generated-token feedback, retains intra-frame depth-autoregressive (depth-AR) residual decoding, and uses residual-token recurrence to provide cross-frame acoustic context only for residual decoding. On 400 bilingual URGENT 2025 utterances, an aligned ablation shows 45.0% lower macro-averaged automatic speech recognition error without temporal feedback, while a double-blind listening test favors depth-AR over fully NAR. FactorGSE outperforms all evaluated streaming baselines in objective speech quality, recognition-based intelligibility, and speaker similarity. An edge implementation runs in real time on a 40-TOPS-class NPU at a 186.7-ms input-context budget. Audio samples and a real-time demo: <https://delaygse.github.io/factorgse/>.

Index Terms— streaming generative speech enhancement, neural audio codecs, selective autoregression

1. INTRODUCTION

Streaming speech enhancement for remote conferencing must preserve natural speech and speaker identity with limited future context. Discriminative systems map noisy input directly to clean speech or enhancement masks, typically by minimizing reconstruction errors against paired clean-speech targets [1, 2, 3]. Generative methods model clean-speech distributions conditioned on degraded input, using continuous representations [4, 5, 6] or discrete codec tokens [7, 8, 9]. Flow and diffusion bridges in continuous space, trained on

paired data with data-prediction losses, retain predictive behavior; their generative formulation alone does not guarantee outperforming comparable discriminative models [10]. Representation choice also matters: continuous-latent DiTSE reports lower perceptual quality than discrete-token Genhancer on DEMO and VBD, while post-hoc codec quantization improves its quality [11], motivating discrete modeling.

However, codec-based generative speech enhancement does not inherently support streaming. Existing offline token-based systems rely on utterance-level conditioning or noncausal sequence processing [7, 8, 9, 12, 13], limiting direct frame-wise generation from bounded input context. A streaming approach uses temporal autoregression (AR) to predict clean tokens from noisy input and previously generated tokens [14]. This history provides context, but errors can propagate into subsequent predictions. Recent non-autoregressive (NAR) approaches avoid temporal generated-token feedback [15, 16], removing this source of error propagation. Yet, when fully NAR prediction treats residual vector quantization (RVQ) codebooks as independent outputs, it omits their explicit coarse-to-fine conditioning. Genhancer’s Parallel variant retains this ordering by generating each quantizer’s full sequence in parallel before advancing to the next [8]. Its comparison uses noncausal input conditioning and does not isolate the effect of frame-local ordering under bounded streaming context. Such frame-local ordering needs no additional input look-ahead, allowing temporal NAR prediction alongside ordered RVQ decoding.

We introduce FactorGSE (Fig. 1), a streaming enhancement framework that uses generated-token context only for residual prediction. Building on Mimi’s semantic-acoustic codec design [17], we predict the semantically distilled first-codebook token from bounded degraded-speech context without generated-token feedback (time-NAR). Residual codebooks are decoded sequentially within each frame (depth-AR), also using cross-frame residual-token history.

Our contributions are:

- A selective streaming factorization combines time-NAR first-codebook prediction with intra-frame depth-AR decoding and residual-only temporal feedback.

*Xin Yuan and Junling Lv contributed equally to this work. †Yanqiang Lei is the corresponding author.

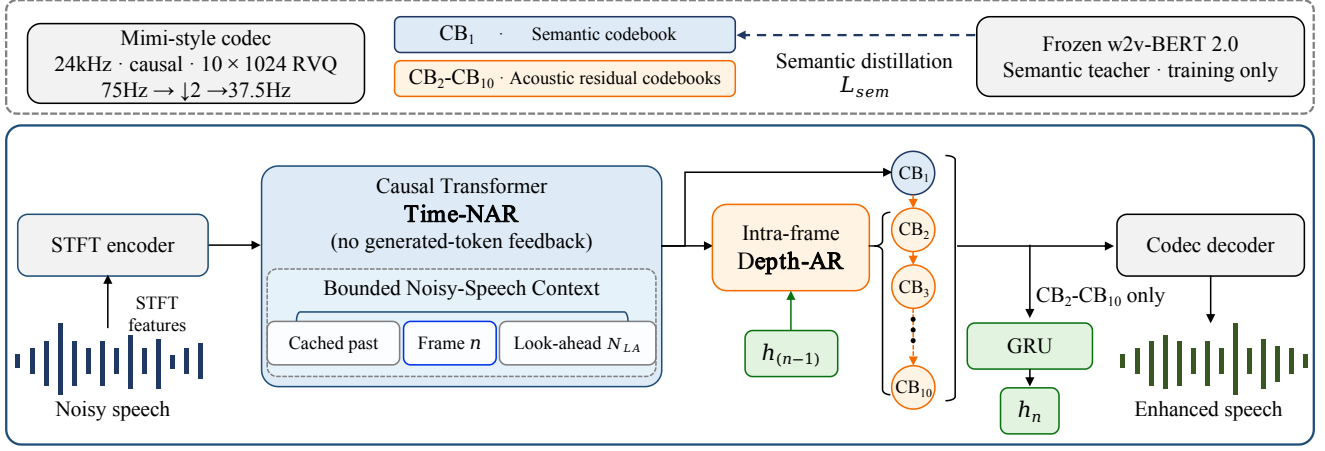


Fig. 1. Overview of FactorGSE. Top (codec pretraining only): semantics from frozen w2v-BERT 2.0 are distilled into CB_1 . Bottom (streaming factorized generator): time-NAR predicts CB_1 from bounded noisy-speech context without generated-token feedback, while depth-AR generates CB_2 – CB_{10} sequentially within each frame. The GRU state h_{n-1} supplies the gated, normalized condition c_n only to the residual decoder; current residual tokens update the state to h_n .

- Aligned controls show 45.0% lower macro-averaged ASR error with fully NAR than time-AR; double-blind listening favors depth-AR over fully NAR.
- On URGENT 2025 [18], FactorGSE outperforms evaluated streaming baselines in DNSMOS, ASR error, and speaker similarity; its compact variant runs in real time on an NPU.

2. PROPOSED METHOD

2.1. Selective Autoregressive Factorization

FactorGSE separates the two dependency axes of an RVQ stream: the first codebook is predicted without temporal target-token feedback, whereas the remaining codebooks retain ordered within-frame prediction and residual-only cross-frame state. Let x and y denote a degraded waveform and its time-aligned clean target. At 37.5 frames/s, the pretrained codec represents clean frame n as $\mathbf{z}_n = (z_n^{(1)}, \dots, z_n^{(K)})$ with $K = 10$ and frame duration $\Delta = 26.67$ ms. A causal front end maps bounded context from x to $\mathbf{s}_n$.

Let $\mathbf{c}_n = C_\psi(\mathbf{z}_{<n}^{(2:K)})$ summarize only residual tokens from preceding frames and let $\mathbf{r}_n = \mathbf{s}_n + \mathbf{c}_n$. For $\mathbf{Z} = \{\mathbf{z}_n\}_{n=1}^T$ and $\Theta = \{\theta_s, \theta_d, \psi\}$, the factorization is

$$p_\Theta(\mathbf{Z} | x) = \prod_{n=1}^T \left[p_{\theta_s}(z_n^{(1)} | \mathbf{s}_n) \times \prod_{k=2}^K p_{\theta_d}(z_n^{(k)} | \mathbf{r}_n, \mathbf{z}_n^{(<k)}) \right]. \quad (1)$$

The first factor is time-NAR: neither generated semantic history $\mathbf{z}_{<n}^{(1)}$ nor residual history $\mathbf{c}_n$ enters the prediction of $z_n^{(1)}$. The second factor is intra-frame depth-AR: it preserves the

coarse-to-fine RVQ order, while cross-frame generated-token recurrence remains confined to residual acoustics. All K tokens share frame index n ; depth recursion therefore adds serial computation but no input-frame look-ahead. The codec decoder renders each completed tuple, so inference emits one 26.67-ms codec frame per call (Fig. 1).

2.2. Semantic Codec and Time-NAR Prediction

We use a causal semantic RVQ codec adapted from the Fish Audio S2-style codec design [19] for 24-kHz speech. A 320-sample internal hop followed by $2 \times$ temporal downsampling yields a 37.5-Hz frame rate; each of the ten codebooks has 1024 entries. To encourage CB_1 to serve as a content anchor, we follow S2’s semantic distillation design: an auxiliary head predicts the frozen w2v-BERT 2.0 teacher’s 16th-layer features from the first codebook’s quantized representation [19, 20]. The semantic feature-regression objective is used only during codec pretraining; the teacher is absent from enhancement training and inference.

For enhancement, a 1024-point STFT (320-sample hop, Hann window) produces 75-Hz degraded-speech features. The acoustic encoder downsamples these by a factor of two to obtain codec-aligned features $\{\mathbf{u}_i\}$ at 37.5 Hz. For frame n , the temporal backbone processes features through $n + N_{LA}$ and outputs $\mathbf{s}_n$, where N_{LA} counts strictly future codec frames. The unsuffixed model’s backbone is initialized from Qwen2-0.5B and uses causal sliding-window attention with ALiBi and a history window of $W = 128$ frames [21, 22]. The input–target shift sets the future codec-frame context. Generated CB_1 tokens are never appended to the backbone input, so its KV cache contains degraded-speech evidence only. This absence of temporal target-token feedback is precisely what time-NAR denotes.

2.3. Intra-Frame Depth-AR Decoding

The residual predictor completes the remaining codebooks of each frame before the codec synthesizes its waveform. Two causally masked Transformer layers (width 896, eight heads, $4H$ feed-forward dimension) condition every depth step on $\mathbf{r}_n$ while exposing only the same-frame prefix $\mathbf{z}_n^{(<k)}$. Thus, $\text{CB}_2\text{--CB}_{10}$ are generated sequentially in RVQ order, but the nine depth steps require no additional input frames.

After each frame, a single-layer 896-dimensional GRU consumes the mean of codebook-position-aware embeddings of the residual tokens. Its layer-normalized output, scaled by a learned sigmoid gate (initialized with $\beta = -2$), forms $\mathbf{c}_{n+1}$; we set $\mathbf{c}_1 = \mathbf{0}$. The update uses clean residual tokens and 0.2 history-context dropout during training, and generated tokens during inference. The state $\mathbf{c}_n$ augments the residual condition $\mathbf{r}_n$, while the first-codebook predictor continues to use $\mathbf{s}_n$.

We train the generator with importance-weighted cross-entropy over the factors in Eq. (1), following DelayGSE’s codebook weighting approach [12]. Each codebook’s prediction loss is weighted according to its contribution to reconstruction quality, recognition, and speaker similarity. Teacher forcing supplies clean same-frame prefixes and clean residual history; inference replaces both with generated tokens.

3. EXPERIMENTS

3.1. Experimental Setup

FactorGSE reuses the DelayGSE training corpus and mixture simulator [12], comprising 30,292 h of clean speech, approximately 590 h of noise, and 700,000 simulated reverberant examples. The factorization variants are trained for 300,000 steps with Adam, a peak learning rate of 5×10^{-4} , 5,000-step warm-up followed by cosine decay, and dynamic batches of approximately 40 s.

We denote the compact configuration by FactorGSE (Small); the unsuffixed FactorGSE is the configuration used for the factorization study and context sweep. Small reduces the temporal backbone from 24 layers of width 896 to 18 layers of width 768 (FFN width 4864 to 3712), and the two-layer depth Transformer from width 896 to 768. Both retain a single-layer residual-history GRU with input and hidden widths matching the backbone: 896 for FactorGSE and 768 for Small. Small also removes the codec’s eight-layer post Transformer and reduces the waveform decoder’s initial channels from 1536 to 1024. Small’s enhancement generator is randomly initialized. Its compact codec is obtained by freezing the pretrained encoder and quantizer and fine-tuning only the reduced decoder. Both configurations use 24-kHz audio and ten codebooks at 37.5 frames/s.

We evaluate on 400 utterances from URGENT 2025 `nonblind_test` [18]: 300 English and 100 Chinese utterances. Both FactorGSE and FactorGSE (Small) use this same

manifest, the same scoring pipeline, and $N_{\text{LA}} = 6$ for the primary comparison. Following the evaluation protocol of DelayGSE, we report DNSMOS P.808 [23, 24] for predicted perceptual quality and use FireRedASR [25] to compute English WER and Chinese CER. Speaker similarity (SIM) is the cosine similarity between ERes2NetV2 embeddings [26] of the enhanced speech and a registered clean reference sample. We summarize recognition results with the equal-language error $\text{Macro ASR} = 100(\text{WER}_{\text{EN}} + \text{CER}_{\text{ZH}})/2$. The Chinese term remains CER; this metric is not a pooled word-level rate. ASR error provides a common-backend comparison of recognition behavior across systems, complementing the perceptual and speaker-similarity measures. Aggregate objective results are fixed-checkpoint point estimates.

All newly evaluated systems share the same manifest and scoring pipeline. We use released checkpoints for UniSE, Stream.FM, FastEnhancer-L, and DeepFilterNet3; Bridge is retrained from public code on internal data [10], while DelayGSE quality scores are inherited from an internal record. Using public checkpoints gives practical reference systems, but their training data, optimization, and native latency definitions differ. The main table therefore compares available systems under a common evaluation protocol; architectural effects are examined with the ablations, which match the manifest, evaluators, training recipe, and effective context budget.

3.2. Ablation of Selective Autoregression

All variants in Table 1 use the primary 186.7-ms effective input-context budget and share the evaluation and training recipe. The two all-codebook time-AR controls predict every codebook autoregressively across time. At each decoding step, their codebook heads emit tokens in parallel, conditioned on generated-token history. The delayed-grid control assigns codebook-dependent temporal offsets to the targets; the aligned control predicts all codebooks for the same physical frame. Thus, parallel heads within a time step do not make either control non-autoregressive.

The fully NAR variant removes generated-token feedback and predicts the frame-aligned codebooks in parallel. The next variant introduces sequential intra-frame residual prediction while retaining time-NAR content prediction. Finally, FactorGSE adds the residual-history GRU. This sequence of comparisons separates target scheduling, temporal feedback, and residual decoding.

The two time-AR controls differ in target scheduling and neither uses the additional residual-history GRU. Frame alignment gives mixed changes (3.50 to 3.48 DNSMOS, 48.80% to 48.40% Macro ASR, and 0.588 to 0.504 SIM), so the delayed-grid row is a scheduling reference rather than temporal-feedback evidence.

Comparing aligned all-codebook time-AR with fully NAR holds target alignment, parallel output heads, and the absence of the residual GRU fixed. Removing temporal

Table 1. Factorization study at a 186.7-ms context budget. Both time-AR controls autoregress across time for all codebooks and use parallel output heads; [†] marks delayed-grid targets. Subsequent frame-aligned variants remove temporal feedback, add intra-frame depth-AR, and add residual history. **Bold marks the best result.**

Variant	DNSMOS $\uparrow$	Macro ASR (%) $\downarrow$	SIM $\uparrow$
All-codebook time-AR (delayed) [†]	3.50	48.80	0.588
All-codebook time-AR (aligned)	3.48	48.40	0.504
Fully NAR	3.58	26.60	0.677
Time-NAR + depth-AR	3.57	26.05	0.696
FactorGSE (+ residual GRU)	3.62	25.45	0.708

Table 2. History diagnostic using one aligned time-AR checkpoint. Only past CB₁ feedback is replaced in the oracle condition; all waveform-synthesis tokens remain predicted. Oracle history is a privileged diagnostic condition.

History condition	DNSMOS $\uparrow$	Macro ASR (%) $\downarrow$	SIM $\uparrow$
Generated history	3.48	48.40	0.504
Oracle CB ₁ history	3.54	28.50	0.641

generated-token feedback reduces Macro ASR from 48.40% to 26.60% (45.0% relative), raises DNSMOS from 3.48 to 3.58, and raises SIM from 0.504 to 0.677. This is the largest recognition change along the ablation sequence. With time-NAR fixed, depth-AR changes the objective scores more modestly: Macro ASR decreases by 0.55 points and SIM rises by 0.019, while DNSMOS changes from 3.58 to 3.57. Adding the residual-history GRU further improves DNSMOS by 0.05, Macro ASR by 0.60 points, and SIM by 0.012. We next examine history sensitivity and perceived quality to understand these different effects.

3.3. Sensitivity to First-Codebook History

To test whether errors in semantic-token history affect the AR control, we evaluate the same aligned all-codebook time-AR checkpoint with *oracle* CB₁ *history*. The clean reference waveform for each test item is encoded and quantized by FishDAC to obtain reference tokens. At frame n , only the past CB₁ tokens used for temporal feedback are replaced by these clean tokens. Past CB₂–CB₁₀ feedback remains generated, and all current tokens are predicted. Waveform synthesis uses the predicted tokens for every codebook, including CB₁; clean tokens are used solely in the history condition. Both conditions use the same 400-utterance evaluation protocol and 186.7-ms input-context budget.

Table 2 shows that correcting only CB₁ history lowers Macro ASR by 19.90 percentage points (41.1% relative), with English WER of 29.8% and Chinese CER of 27.2%. DNSMOS and SIM also improve. Because the synthesized CB₁ tokens remain predicted, this gain comes from chang-

Table 3. Double-blind overall-quality preferences: fully NAR versus time-NAR + depth-AR, both without GRU. Percentages use all 160 judgments, including ties. Results are descriptive, with repeated listeners and clips.

Preference	Judgments	Share (%)
Depth-AR preferred	103	64.4
Tie	42	26.3
Fully NAR preferred	15	9.4

ing the history condition rather than inserting clean tokens into the output. The result supports sensitivity to errors in semantic-token history in this AR control, motivating the time-NAR content predictor. It does not isolate every source of the AR/NAR difference, and oracle history is unavailable at deployment.

3.4. Overall-Quality Listening Evaluation

Informal listening reveals tremulous artifacts in fully NAR outputs even when DNSMOS is high. We therefore compare fully NAR with time-NAR + depth-AR in a double-blind paired test of *overall speech quality*, at $N_{LA} = 6$ and without the residual GRU in either system. Each of eight listeners rates 20 clips randomly selected from the 400-utterance test set, yielding 160 judgments over 119 unique clips. Listeners cover the languages of their assigned material. Presentation order is varied. The five response levels indicate that A is much better, better, equal, worse, or much worse than B; we combine the two preference strengths on each side for reporting.

Despite nearly equal DNSMOS (3.58 for fully NAR and 3.57 for depth-AR), listeners favor depth-AR in 103 judgments against 15, with 42 ties (Table 3). The preference direction holds in both display positions and for seven of eight listeners. These judgments support a perceptual benefit from frame-local codebook conditioning that the aggregate DNSMOS scores do not capture. The test measures overall quality; the tremulous artifacts motivate the comparison but are not scored separately.

3.5. Comparison with Available Enhancement Systems

At the primary $N_{LA} = 6$ setting, FactorGSE obtains 3.62 DNSMOS, 25.45% Macro ASR, and 0.708 SIM (Table 4), giving the lowest Macro ASR point estimate among the evaluated streaming systems. External systems retain their native configurations, so this comparison locates the method among available systems rather than isolating architectural effects under matched training data and end-to-end latency.

At the same input-context budget, FactorGSE (Small) obtains 3.63 DNSMOS, 25.75% Macro ASR (26.7% English WER and 24.8% Chinese CER), and 0.729 SIM. It retains comparable objective quality while reducing the full inference parameter count from 578.45M to 255.29M and MACs

Table 4. Comparison on URGENT 2025. Both FactorGSE configurations use 186.7-ms input context; other systems retain their own configurations. Bold/underlined scores mark the best/second-best streaming results; [†] denotes inherited quality scores. Parameter counts and MACs use a common profiling protocol; the FactorGSE GRU contribution is calculated analytically. MACs are normalized per second of input audio for 10-s, 24-kHz inputs at batch size one. FactorGSE complexity includes codec synthesis and GRU. Bridge uses five ODE steps, and Stream.FM uses five sampling steps.

Method	DNSMOS ↑	Macro ASR (%) ↓	SIM ↑	Params (M)	MACs (G/s)
Clean	3.67	4.55	1.000	—	—
Clean codec recon.	3.67	4.90	0.900	—	—
Noisy input	2.74	15.55	0.697	—	—
<i>Offline enhancement</i>					
UniSE [13]	3.85	33.25	0.660	253.98	42.78
DelayGSE [†] [12]	3.77	16.95	0.634	1286.32	259.05
Bridge [10]	3.37	33.20	0.593	2.16	182.71
<i>Streaming enhancement</i>					
Stream.FM [27]	3.24	43.60	0.660	52.51	705.22
FastEnhancer-L [28]	3.23	33.90	0.658	1.14	32.38
DeepFilterNet3 [29]	3.32	28.40	0.699	2.14	0.33
FactorGSE	<u>3.62</u>	25.45	<u>0.708</u>	578.45	68.89
FactorGSE (Small)	3.63	<u>25.75</u>	0.729	255.29	32.65

EN WER / ZH CER (%): Clean 5.5 / 3.6; codec reconstruction 5.8 / 4.0.

from 68.89 to 32.65 G/s. The reduction spans the temporal backbone, depth decoder, and codec decoder; Small gives the highest DNSMOS and SIM point estimates among these streaming systems, with Macro ASR 0.30 points above the larger configuration.

The offline rows provide additional quality references: UniSE has higher DNSMOS, while DelayGSE has higher DNSMOS and lower Macro ASR but lower SIM. Noisy input retains the lowest Macro ASR (15.55%). Perceptual improvement and recognition accuracy need not move together: resynthesis can make speech sound fuller and cleaner while introducing signal distortions that affect an ASR backend. ASR therefore compares recognition behavior under a fixed recognizer, complementing DNSMOS, speaker similarity, and the overall-quality listening test. The difference from noisy input remains part of this quality–recognition trade-off.

3.6. Input Context and Deployment Reference

For codec-frame duration $\Delta = 26.67$ ms, we define the input-context budget as $L_{\text{ctx}} = (N_{\text{LA}} + 1)\Delta$, excluding processing and audio I/O. Our remote-conferencing application targets an input-context budget below 200 ms. We choose $N_{\text{LA}} = 6$, giving 186.7 ms, as the largest integer look-ahead setting within that budget; $N_{\text{LA}} = 7$ would require 213.3 ms.

The four sweep configurations in Fig. 2 are trained separately for 300,000 steps each, using their respective look-ahead settings and the same training recipe. They compare

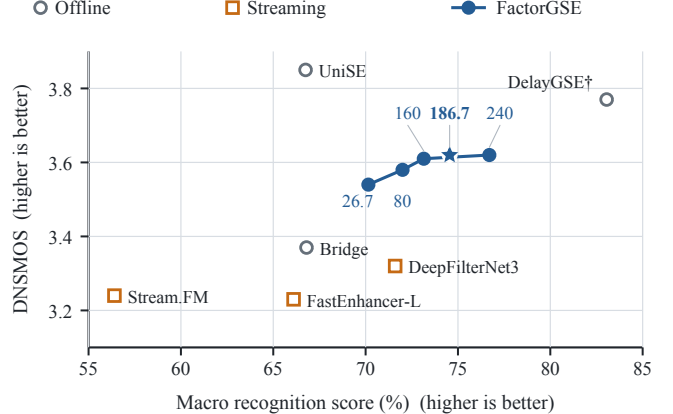


Fig. 2. DNSMOS–recognition trade-off on URGENT 2025 (upper right is better). Macro recognition score is 100–Macro ASR error, not token accuracy. Blue labels denote input context in ms. Connected points represent separately trained 300,000-step configurations; the star is the separate 186.7-ms primary run. Noisy input is omitted; [†] marks an inherited record.

fixed-context models, not inference-time adjustments to one checkpoint. The 186.7-ms primary configuration is a separate training run and is marked with a star. Across the four sweep models, increasing input context from 26.67 to 240 ms raises DNSMOS from 3.54 to 3.62 and reduces Macro ASR from 29.85% to 23.30%, while SIM stays within 0.711–0.716. Additional acoustic context thus improves recognition in this comparison, but the 240-ms configuration exceeds the chosen application budget.

As a deployment reference, the C++ implementation of FactorGSE (Small) reaches RTF 0.81 on an edge device with a 40-TOPS-class NPU at the primary setting. The platform identity is confidential; RTF describes average processing cost, and end-to-end latency is not measured here.

4. CONCLUSION

We presented FactorGSE, which combines time-NAR content prediction with frame-local depth-AR and residual acoustic history for streaming generative speech enhancement. Aligned controls show a large recognition improvement when removing temporal generated-token feedback. Correcting only past CB₁ feedback in the same AR checkpoint substantially improves its output, supporting sensitivity to semantic-history errors. Complementary listening judgments favor depth-AR over fully NAR decoding despite nearly identical DNSMOS. These observations support treating temporal content prediction and within-frame acoustic generation differently. A compact configuration retains comparable objective quality at the selected 186.7-ms input-context budget.

Evaluation uses one 400-utterance bilingual benchmark

and a descriptive listening test. External systems differ in training data and native latency, and the deployment RTF measures average processing rather than end-to-end delay. Broader generalization and statistical uncertainty remain to be assessed.

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